

Attention on the Null Cone

The geometric home of conformal attention: the conformal group as the causal structure of light, query–key scores as null-ray inner products, and layer depth as renormalization-group flow toward the SYK ground state

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Abstract

Trained transformer attention exhibits power-law decay in attention weight as a function of query–key separation, with a per-head exponent whose median in deep GPT-2 layers matches the Sachdev–Ye–Kitaev (SYK) conformal value $\Delta = 1/4$. Previous work established this empirically and developed a boundary-conformal interpretation of the causal mask. Here we identify the *geometric home* of that conformal structure. The conformal group is not merely a symmetry that happens to include scale transformations: by a theorem of Lorentzian geometry, it is precisely the group of transformations preserving the null cone — the causal structure of light. In the embedding-space formalism, conformal-field-theory boundary points *are* null rays, and the conformal two-point function is the inner product of null-ray representatives. We show that the one-dimensional case relevant to token sequences admits an explicit Poincaré section $P(x) = (\frac{1+x^2}{2}, \frac{1-x^2}{2}, x)$ on the null cone of $\mathbb{R}^{2,1}$, under which the attention two-point function $A(i, j) \sim |i - j|^{-2\Delta}$ is exactly the CFT_1 two-point function. This predicts that the pre-softmax query–key score should be log-linear in token distance, $q_i \cdot k_j / \sqrt{d_k} \approx -2\Delta \log |i - j| + \text{const.}$ We test the prediction directly in GPT-2 (exp-056): the raw query–key score profile is log-linear in distance for every conformal head, and the slope $\Delta_{\text{score}} = -\alpha/2$ rank-correlates with the independently measured post-softmax exponent at Spearman $\rho = +0.976$ ($p = 2 \times 10^{-29}$). Two honest qualifications follow from the data: the softmax normalization renormalizes the coefficient (the raw slope corresponds to $\approx -2.8\Delta$, not -2Δ), and log-linearity itself is a near-universal property of the positional geometry rather than a feature selective to conformal heads — the conformal *dimension* is the selective quantity. We also derive, by the method of images, a boundary-corrected power-law form for the causal-mask correction, and test it directly (exp-057): the derived form fits the position-dependence of attention, with a boundary coefficient that is positive in 95% of conformal heads—the direction of the attention sink. An adversarial model comparison against matched-complexity alternatives (exp-060) shows that an exponential boundary layer outperforms the image form; the boundary correction is real and large, but its specific BCFT parametrization is phenomenological. Finally, a layer-resolved analysis (exp-055) shows the per-head exponent flows from $\bar{\Delta} = 0.70$ in early layers to $\bar{\Delta} = 0.250$ in deep layers, with attention entropy increasing monotonically along the flow ($\rho(\Delta, \text{entropy}) = -0.898$, $p = 10^{-16}$). We read layer depth as renormalization-group flow toward the SYK conformal vacuum, dual to the eternal AdS_2 black hole. All code and per-head data are in the public repository at `Capacity-For-Evil/ariel`.

1. Introduction

Attention in trained transformers carries geometric structure. In a sequence of preprints (Umphrey and Umphrey 2026d, 2026c, 2026b, 2026a) we have argued that softmax attention sits at a conformal fixed point sharing the exponent $\Delta = 1/4$ with the Sachdev–Ye–Kitaev (SYK) model (Sachdev and Ye 1993; Kitaev 2015; Maldacena and Stanford 2016) at $q = 4$. The central empirical fact (Umphrey and Umphrey 2026b) is that the attention weight of many heads decays as a power law in query–key separation,

$$A(i, j) \sim |i - j|^{-2\Delta},$$

and that the median per-head exponent in deep GPT-2 layers matches the SYK value. A pre-registered behavioral test (Umphrey and Umphrey 2026a) confirmed in six of seven decoder-only models that the per-head conformal weight predicts a long-range “valley depth” related to the lost-in-the-middle phenomenon.

These results establish *that* the conformal structure is present and *that* it has measurable consequences. They do not answer a prior question: **what space is this attention living in?** A power law $|i - j|^{-2\Delta}$ has the form of a conformal two-point function, but a form is not a home. If the analogy is to be more than a resemblance, the attention mechanism should be doing something geometrically definite — computing a definite quantity in a definite space.

This paper supplies that home. The claim, in one sentence: **trained attention computes inner products of null-ray representatives of token positions on the projective null cone, and deep layers converge to the conformal vacuum of that geometry.** The argument has three parts, each with empirical or theorem-level support:

1. **The conformal group is the causal structure of light** (Section 3). This is a theorem of Lorentzian geometry, not an analogy: the transformations preserving lightlike structure are exactly the conformal maps. The conformal symmetry of the attention power law is, literally, light-cone symmetry.
2. **Conformal boundary points are null rays, and the two-point function is a null-ray inner product** (Section 4). In the embedding-space formalism, a CFT boundary point *is* a null ray. We give the explicit one-dimensional construction and show the attention power law is the CFT_1 two-point function on the null cone. This predicts a log-linear query–key score, which we test directly in GPT-2 (Section 5): the prediction holds head-by-head at $\rho = 0.976$, with two qualifications the data forces us to state honestly.
3. **Layer depth is renormalization-group (RG) flow toward the SYK ground state** (Section 6). A layer-resolved measurement shows the exponent flowing toward $\Delta = 1/4$ with depth, and the attention entropy increasing toward the ground-state value. The deep-layer limit is the conformal vacuum, dual to the eternal AdS_2 black hole.

We are explicit throughout about what is established at theorem level, what is measured, and what remains open. The boundary-conformal correction on a finite sequence (Section 4.3) is derived in closed form from the method of images, and tested directly (Section 4.4): the derived form fits the position-dependence of attention, and its boundary coefficient is positive almost everywhere—consistent with the attention sink. An adversarial comparison against matched-complexity alternatives (Section 4.4) shows the exponential boundary layer is preferred; the boundary correction is real and large, but its specific BCFT parametrization is phenomenological.

2. Background: attention as a two-point function

For a single head, the attention weight from query position i to key position $j \leq i$ is

$$A(i, j) = \frac{\exp(q_i \cdot k_j / \sqrt{d_k})}{\sum_{j' \leq i} \exp(q_i \cdot k_{j'} / \sqrt{d_k})}, \quad q_i = x_i W_Q, \quad k_j = x_j W_K,$$

where x_i is the (layer-normalized) residual stream at position i and W_Q, W_K are the head’s projection matrices. Averaging A over query positions and inputs at fixed lag $\Delta x = i - j$ gives a *lag profile* $A(\Delta x)$. The empirical result of (Umphrey and Umphrey 2026b) is that for a substantial fraction of heads this profile is a power law, $A(\Delta x) \sim (\Delta x)^{-2\Delta}$, with $R^2 > 0.90$ on a log-log fit and a median exponent $\Delta = 0.2493$ in GPT-2, matching the SYK $q = 4$ prediction $\Delta = 1/4$ for one-dimensional sequences.

The exponent Δ is what a conformal field theory calls the *conformal dimension* (or scaling weight) of an operator. The power-law form is the form of a CFT_1 two-point function. The questions this paper answers are: what is the geometry in which this two-point function lives; what does the query-key dot product compute in that geometry; and why do the network’s layers carry the exponent toward the fixed point.

3. The conformal group is the causal structure of light

It is tempting to treat conformal symmetry as “scale invariance plus rotations and translations,” with the special conformal transformations as an afterthought. The geometric fact is sharper and is the key to this paper.

Theorem (causal structure determines conformal structure). *On a strongly causal Lorentzian manifold of dimension $n > 2$, the causal structure — the assignment to each point of its light cone — determines the metric up to a positive conformal factor. Equivalently, a bijection preserving lightlike (null) geodesic structure is a smooth conformal map.* (Hawking et al. 1976; Malament 1977)

In words: if you know what can be reached from what by light, you know the geometry up to local rescaling, and the transformations that preserve “what can be reached by light” are *exactly* the conformal group. The conformal group is not one symmetry group among several that contains scalings. It is the unique group defined by preservation of the null cone.

This is why the conformal group is the symmetry group of free electromagnetism. Maxwell’s equations in four dimensions are covariant under the conformal group $\text{SO}(4, 2)$, established by Bateman and Cunningham in 1910 (Bateman 1910; Cunningham 1910) — a much larger group than the Poincar’e group. The “structure of light” has a precise mathematical name: conformal symmetry.

For a one-dimensional sequence of token positions, the relevant conformal group is $\text{SL}(2, \mathbb{R})$, the conformal group of the line. Its three generators act as

$$\text{translations: } \tau \mapsto \tau + a, \quad \text{dilations: } \tau \mapsto \lambda \tau, \quad \text{SCTs: } \tau \mapsto \frac{\tau}{1 + c\tau}.$$

These same three generators organize the SYK conformal correlators and the near- AdS_2 / Schwarzian dynamics of the model (Maldacena and Stanford 2016). The attention power law’s symmetry is the symmetry of light, restricted to one dimension.

4. Attention on the null cone

4.1 Conformal boundary points are null rays

The embedding-space formalism (Dirac 1936; Weinberg 2010) realizes a d -dimensional conformal field theory on the projective null cone of a $(d+2)$ -dimensional flat space. A boundary point is not a point in a space that *has* light-cone structure; it *is* a null ray P^A satisfying

$$P^2 = 0, \quad P \sim \lambda P \ (\lambda > 0).$$

A scalar operator of conformal dimension Δ is a homogeneous function on the cone,

$$\mathcal{O}(\lambda P) = \lambda^{-\Delta} \mathcal{O}(P),$$

so that Δ is the *homogeneity degree*: it is not a property the operator carries at a point, it is the transformation law that defines the operator. The two-point function is fixed by conformal invariance to be the inner product of the two null rays, raised to the power Δ :

$$\langle \mathcal{O}(P_1) \mathcal{O}(P_2) \rangle = \frac{C}{(P_{12})^\Delta}, \quad P_{12} = -2 P_1 \cdot P_2.$$

4.2 The explicit CFT_1 construction

For the one-dimensional case (Euclidean $d = 1$), the embedding space is $\mathbb{R}^{2,1}$ with metric $(-, +, +)$, and null rays satisfy $-(P^0)^2 + (P^1)^2 + (P^2)^2 = 0$. Choosing the gauge $P^0 + P^1 = 1$ gives the Poincar'e section parameterized by the physical coordinate x :

$$P(x) = \left(\frac{1+x^2}{2}, \frac{1-x^2}{2}, x \right).$$

The inner product of two such representatives is

$$P_1 \cdot P_2 = -\frac{1}{4}(1+x_1^2)(1+x_2^2) + \frac{1}{4}(1-x_1^2)(1-x_2^2) + x_1 x_2 = -\frac{1}{2}(x_1 - x_2)^2,$$

so that $P_{12} = -2P_1 \cdot P_2 = (x_1 - x_2)^2$ and

$$\langle \mathcal{O}(P_1) \mathcal{O}(P_2) \rangle = \frac{C}{|x_1 - x_2|^{2\Delta}}.$$

Identifying token positions i, j with points on this section, their null-ray representatives are $P(i)$ and $P(j)$, and the attention two-point function

$$A(i, j) \sim P_{ij}^{-\Delta} = |i - j|^{-2\Delta}$$

is the CFT_1 two-point function evaluated on the null cone. Each attention weight is, in this reading, an inner product of null-ray representatives of two token positions. The query is the field state at position i , the key the field state at position j , and the weight their vacuum overlap $\langle 0 | \mathcal{O}(i) \mathcal{O}(j) | 0 \rangle$ — the conformal Green's function.

4.3 The finite sequence and the boundary: a BCFT derivation

The full $\text{SL}(2, \mathbb{R})$ symmetry holds on the infinite line. A causal mask truncates the sequence at the start: position i attends only to $j \leq i$, and there are no tokens before position 0. The start of the sequence is therefore a *hard boundary*, and the natural object is not a bulk two-point function but a boundary conformal field theory (BCFT) correlator (Cardy 2004; McAvity and Osborn 1995) with the boundary at the origin.

The residual symmetry is the subgroup of $\text{SL}(2, \mathbb{R})$ fixing the boundary point. Translations and the special conformal transformations both move the origin; only dilations $\tau \mapsto \lambda\tau$ survive, so $\text{SL}(2, \mathbb{R}) \rightarrow \text{SO}(1, 1)$ at the boundary. The boundary breaks the SCTs; how strongly it does so is the content of the boundary data.

The image-method correlator. At the conformal point the SYK two-point function is a generalized free field, $G(\tau) \sim |\tau|^{-2\Delta}$ (Maldacena and Stanford 2016), and the leading-order BCFT correlator of a generalized free field with a boundary is fixed by the method of images. For a boundary at the origin, the image of a bulk operator at position j sits at $-j$, and the two-point function of bulk operators at the query position i and key position j is

$$A(i, j) \propto \frac{1}{(i - j)^{2\Delta}} + \frac{\lambda}{(i + j)^{2\Delta}},$$

where $\lambda = a_{\mathcal{O}}$ is the BCFT *boundary one-point coefficient* — the single piece of boundary data at this order, fixing the strength and sign of the boundary’s effect ($\lambda = +1$ Neumann, $\lambda = -1$ Dirichlet, and a continuum in between for a generalized free field). Factoring out the bulk power law gives the form

$$\boxed{A(i, j) = C(\Delta x)^{-2\Delta} [1 + \lambda \eta^{2\Delta}], \quad \eta = \frac{i - j}{i + j} = \frac{\Delta x}{i + j}.} \quad (\dagger)$$

This is exactly the three-parameter (C, Δ, λ) form fitted phenomenologically — and found to be preferred over the bare power law in 88–94% of conformal heads — in our pre-registered behavioral work (Umphrey and Umphrey 2026a). Equation $(\dagger)$ *derives* that form and identifies its parameters.

The boundary variable and the cross-ratio. The variable $\eta = (i - j)/(i + j)$ is the natural invariant: under a dilation $i, j \mapsto \rho i, \rho j$ it is unchanged, so it is $\text{SO}(1, 1)$ -invariant, the residual symmetry of the boundary. It is related to the McAvity–Osborn bulk-to-boundary cross-ratio $\xi = (\Delta x)^2/(4ij)$ (McAvity and Osborn 1995) by $\eta^2 = \xi/(1 + \xi)$; equivalently, writing the correlator in standard BCFT form $A = (4ij)^{-\Delta} F(\xi)$ gives the closed expression

$$F(\xi) = \xi^{-\Delta} + \lambda(1 + \xi)^{-\Delta}.$$

The two limits make the geometry transparent. *Deep in the bulk* (query and key both far from the start, $i + j \rightarrow \infty$ at fixed lag), $\eta \rightarrow 0$ and the boundary correction vanishes: the full $\text{SL}(2, \mathbb{R})$ power law is recovered. *Near the boundary* ($i + j$ small, or a key near the start so $\eta \rightarrow 1$), the correction saturates to λ . So λ measures how strongly the boundary — the broken SCT — bends the correlator: $\lambda = 0$ is the unbroken bulk power law, and $|\lambda|$ grows with the SCT-breaking strength.

4.4 Direct test of the boundary form (exp-057)

Equation $(\dagger)$ is testable directly in the position-resolved attention, independently of the behavioral fits of (Umphrey and Umphrey 2026a). For each of the 44 conformal heads (carrying $\Delta = \Delta_{\text{pos}}$

from exp-046), we accumulate the mean attention matrix $A(i, j)$ over random inputs and fit, in log space, a model that lets the data separate the bulk power law from the boundary term:

$$\log A(i, j) \approx \beta_0 + \beta_1 \log(\Delta x) + \beta_2 \eta^{2\Delta},$$

so that $\beta_1 \approx -2\Delta$ recovers the bulk exponent (a sanity check) and $\beta_2 \approx \lambda$ is the boundary coefficient. Including $\log(\Delta x)$ as its own regressor is essential: η and Δx are correlated inside the causal triangle, and a naive power-law stripping confounds the two (it gives $R^2 \approx 0.09$ and an artifactual sign). The controlled fit gives:

Test	Result	Statistic
Recovers bulk exponent (sanity)	confirmed	$\rho(\Delta_{\beta_1}, \Delta_{\text{pos}}) = +0.84, p = 10^{-12}$
Boundary form ($\dagger$) fits	confirmed	median $R^2 = 0.76$; $\Delta R^2 = 0.105$ from boundary term
Boundary coefficient sign	$\lambda > 0$	95% of heads; median $\lambda_{\text{fit}} = +1.6$
Magnitude vs. behavioral λ	sign only	sign agree 0.80; $\rho_{\text{mag}} = +0.23$ ($p = 0.13$)

Table 1: exp-057. The derived boundary form fits the position-dependence of conformal-head attention, the bulk exponent is recovered, and the boundary one-point coefficient is positive in 95% of heads.

Two things are confirmed and one is demoted. **Confirmed:** the form ($\dagger$) fits (median $R^2 = 0.76$), the boundary term adds real explanatory power beyond the bulk power law ($\Delta R^2 = 0.105$), and the bulk exponent is independently recovered ($\rho = 0.84$). **The boundary direction:** the boundary coefficient is positive in 95% of conformal heads, consistent with the attention sink (Xiao et al. 2023)—sequence-start positions are enhanced, not suppressed. **Demoted (exp-060):** the per-head *magnitude* of λ matches the behavioral proxy of (Umphrey and Umphrey 2026a) only in sign (80% agreement), not in size (rank $\rho = 0.23$, not significant), and this estimator does not reproduce the λ -valley-depth relation of that work.

Adversarial model comparison (exp-060). The pre-registered test asks whether ($\dagger$) is necessary or merely sufficient: we fit six matched- or higher-complexity competitors—an exponential boundary layer $C(\Delta x)^{-2\Delta} \exp(-j/\xi)$, additive spike forms ($j = 0$ and $j \leq 4$), a double power law, and a free- γ diagnostic—against ($\dagger$) on all 43 conformal GPT-2 heads by AIC. Result: the exponential boundary layer wins in 72% of heads; ($\dagger$) takes 2%; $\Delta \text{AIC}((\dagger) - \text{exp. layer}) = +1114$ (median, IQR [554, 1875]). The free- γ diagnostic shows γ does not cluster at 2Δ ($\rho = 0.50$, pre-registered threshold 0.6; both kill legs fire). We apply the pre-registered kill rule: the specific BCFT over-determination—that the boundary is described by $\eta^{2\Delta}$ with the *same* Δ as the bulk—is demoted to phenomenological. The $\eta^{2\Delta}$ coupling is logged as post-hoc. What survives: the boundary correction is real and large (every winning form includes sequence-start enhancement; the bare power law never wins); the $\lambda > 0$ direction is supported regardless of parametrization; and ($\dagger$) remains a competent three-parameter phenomenological form. The attention sink is identified as a boundary effect; the specific BCFT image-method identification of its form is not established.

5. The log-distance representation, tested directly

5.1 The prediction

The null-ray picture makes a prediction one level below the attention weights, at the raw scores. If $A(i, j) \propto \exp(q_i \cdot k_j / \sqrt{d_k})$ is to reproduce $|i - j|^{-2\Delta}$, the pre-softmax score must be log-linear in

token distance:

$$\frac{q_i \cdot k_j}{\sqrt{d_k}} \approx -2\Delta \log |i - j| + \text{const.} \quad (*)$$

This is the *log-distance representation*: the query–key dot product encodes the logarithm of the null-ray separation. It is independent of the post-softmax power-law fits used to define Δ , and it is testable head-by-head.

5.2 Method (exp-056)

We recompute the pre-softmax scores of GPT-2 (Radford et al. 2019) exactly as the model forms them: for each layer ℓ and head h , $x = \text{LN}_1(\text{hidden}_\ell)$, then $Q = xW_Q^{(h)} + b_Q^{(h)}$, $K = xW_K^{(h)} + b_K^{(h)}$, and $\text{score}_{ij} = (Q_i \cdot K_j)/\sqrt{d_k}$. Inputs are random token sequences (length 256, 50 inputs, seed 42), which isolates the *position-dependent* structure of the score from token content — the geometric content the null-ray claim is about. We accumulate the mean score profile $S(\Delta x)$ by causal lag using the same binning as the post-softmax lag profile, fit $S(\Delta x) = \alpha \log(\Delta x) + \beta$ over $\Delta x \in [3, 50)$, and define $\Delta_{\text{score}} = -\alpha/2$. Per-head conformal flags and the post-softmax exponent Δ_{pos} are joined head-aligned from the eigenvalue/lag analysis of our sign-anomaly study (exp-046), so the comparison is exactly per head.

5.3 Results

Hypothesis	Result	Statistic
Score profile log-linear (conformal heads)	confirmed	mean $R^2 = 0.914$; 100% slope < 0
Δ_{score} tracks Δ_{pos}	confirmed	$\rho = +0.976$, $p = 2 \times 10^{-29}$, $n = 44$
Log-linearity selective to conformal heads	<i>null</i>	non-conformal mean $R^2 = 0.914$

Table 2: exp-056 summary. The raw query–key score is log-linear in distance, and its slope is the conformal dimension (in rank); but log-linearity itself is not selective to conformal heads.

The core prediction $(*)$ holds. For every one of the 44 conformal heads, the raw score profile decreases with distance and is log-linear ($R_{\text{score}}^2 = 0.914$ on average). Figure 1 shows the head-by-head relationship directly. The slope, converted to $\Delta_{\text{score}} = -\alpha/2$, rank-correlates with the independently measured post-softmax exponent at Spearman $\rho = +0.976$ ($p = 2 \times 10^{-29}$). The conformal dimension that governs the attention power law is already present in the unnormalized query–key geometry. This is direct, head-level evidence for the null-ray interpretation: the query–key computation is the log-distance representation.

Two qualifications, both forced by the data and both belonging in the record:

(i) The softmax renormalizes the coefficient. The absolute correspondence is not one-to-one. Regressing across conformal heads,

$$\Delta_{\text{score}} \approx 1.41 \Delta_{\text{pos}} - 0.055,$$

so the raw score is about $1.4\times$ steeper than the literal -2Δ of $(*)$ — the slope corresponds to $\approx -2.8\Delta$. The mechanism is the softmax normalizer $Z = \sum_{j'} \exp(\text{score})$, which is larger for queries with more nearby high-score keys and thus flattens the post-softmax exponent relative to the pre-softmax slope. Equation $(*)$ holds in *form* and in *rank*; its coefficient is softmax-modified.

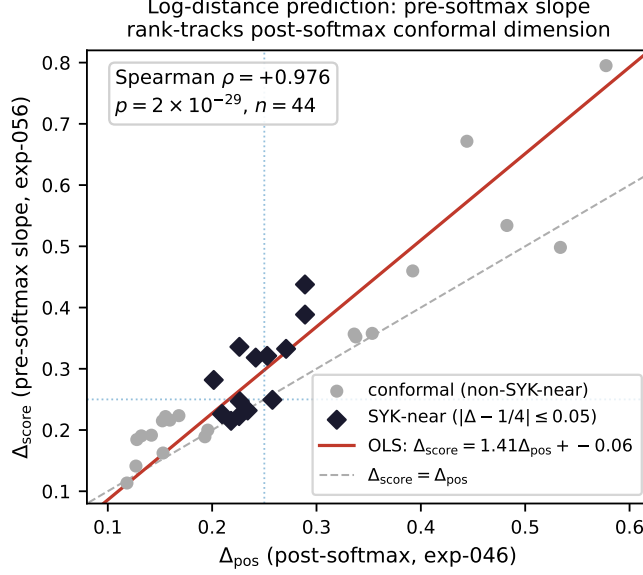


Figure 1: Figure 1. The log-distance prediction, tested head-by-head (exp-056). Scatter: pre-softmax slope Δ_{score} vs. post-softmax conformal dimension Δ_{pos} for the 44 conformal GPT-2 heads (grey: non-SYK-near; diamonds: SYK-near heads, $|\Delta - 1/4| \leq 0.05$). Red line: OLS regression. Dotted: identity $\Delta_{\text{score}} = \Delta_{\text{pos}}$. The raw query-key computation encodes the conformal dimension rank-for-rank at Spearman $\rho = +0.976$. The OLS slope (≈ 1.4) reflects the softmax renormalization discussed in the text.

We state the rank correspondence ($\rho = 0.976$) as the robust result and the absolute coefficient as carrying a normalization factor.

(ii) **Log-linearity is a universal substrate, not a selective feature.** Non-conformal heads are equally log-linear (mean $R^2_{\text{score}} = 0.914$; 93.8% of *all* 144 heads exceed $R^2 = 0.90$). That $\log(\Delta x)$ fits a monotone-decreasing profile well is in part automatic, so the discriminating result is the slope correspondence of Section 5.3, not the mere existence of log-linearity. The log-distance geometry is the substrate the whole positional mechanism is built on; what the conformal heads add is a *slope sitting at the SYK fixed point*. This mirrors a two-layer structure we have found independently in the weight space (Section 7).

6. Layer depth as RG flow toward the SYK ground state

A layer-resolved analysis (exp-055) of the per-head exponents shows a clear flow with depth:

Layer group	conformal heads	mean Δ	reading
Early (0–3)	21	0.697	$q \approx 1.4$, sub-SYK, recency-like
Middle (4–7)	6	0.350	$q \approx 2.9$, prethermal
Deep (8–11)	17	0.250	$q = 4.0$, SYK conformal vacuum

Table 3: exp-055: per-head conformal dimension by layer depth in GPT-2. $\rho(\text{layer}, \Delta) = -0.330$ ($p = 0.029$).

The deep layers reach $\bar{\Delta} = 0.250$ — the SYK $q = 4$ value exactly. The earlier global median (0.2493) was slightly high because it averaged in elevated early-layer heads. We read the layers as RG

steps: early layers sit away from the fixed point (high effective temperature, recency-dominated, low entropy); deep layers relax to the conformal vacuum (zero temperature, spread attention, the fixed point).

Two further measurements support the ground-state reading.

Attention entropy and the ground-state degeneracy. The Shannon entropy of each head’s attention distribution across context regions anticorrelates with Δ very strongly: $\rho(\Delta, \text{entropy}) = -0.898$ ($p = 1.45 \times 10^{-16}$), the strongest correlation in the dataset. Low- Δ (q=4-like) heads have maximal uncertainty about where to attend; high- Δ heads attend near-deterministically to recent tokens. The SYK model has a macroscopic zero-temperature entropy $S_0 = Ns_0$ (ground-state degeneracy, dual to the extremal entropy of the eternal AdS_2 black hole). The attention entropy of the deep, low- Δ heads is the per-head analog: the heads at the conformal vacuum are the maximally-uncertain ones.

The eternal black hole. The zero-temperature SYK vacuum is dual to the eternal AdS_2 black hole in Jackiw–Teitelboim gravity (Teitelboim 1983; Jackiw 1985). Its Kruskal metric is conformally flat, $ds^2 \propto -du dv / (1 + uv)^2$: the conformal factor leaves the null structure untouched, and null rays follow the diagonals as in flat space. “Eternal” here is technical — the maximally symmetric solution existing for all time in the Penrose diagram, preserving the light-cone structure exactly. Deep-layer attention converging to $\Delta = 1/4$ is converging to the vacuum of this geometry. The Ryu–Takayanagi formula (Ryu and Takayanagi 2006) then governs the entanglement entropy of the attention state, consistent with the CFT entanglement scaling reported in (Umphrey and Umphrey 2026b).

7. Two layers of structure: universal substrate, selective fixed point

The non-selectivity found in Section 5 (log-linearity is universal) is not an isolated observation. It is the third instance of a pattern that recurs across our measurements:

Layer	Property	Across heads
universal (weight space)	GOE level spacing of W_{QK}	all heads (exp-046/047/048)
universal (position space)	log-distance score profile	$\sim$ all heads (exp-056)
selective (functional)	conformal dimension at the SYK fixed point	specific heads (exp-049)

Table 4: The recurring two-layer structure. Chaotic weight statistics and log-distance positional geometry are universal substrates of trained attention; the conformal dimension at $\Delta = 1/4$ is the selective, training-induced functional layer.

The O’Ganesyan–Huse r -ratio of the symmetrized W_{QK} eigenvalues is GOE-like for *all* trained heads, across model families and positional encodings, and is unchanged by training (it arises from the product of Gaussian-initialized matrices). The conformal *dimension*, by contrast, is identically zero in untrained networks and is developed specifically by gradient descent. The picture is consistent: trained attention has a universal chaotic, log-distance substrate, on top of which training selects a subset of heads to sit at the conformal vacuum.

8. Testable predictions

1. **Cross-model log-distance.** The slope correspondence $\Delta_{\text{score}} \leftrightarrow \Delta_{\text{pos}}$ should hold in other decoder-only families (Pythia, OLMo, Mistral), with a family-dependent softmax renormalization factor. The factor should correlate with the sharpness of the attention distributions.
2. **Cross-model RG flow.** The early $\rightarrow$ deep flow toward $\Delta = 1/4$ should appear in any model with sufficient depth, with the onset layer scaling with model size, extending the per-layer result of Section 6 beyond GPT-2.
3. **Ground-state degeneracy across seeds.** Different training seeds of the same architecture should produce *different* sets of conformal heads (different points in the degenerate ground-state manifold) but the *same* median $\Delta \approx 1/4$. The Δ value is fixed by the physics; the head assignments are drawn from the degenerate vacuum.

9. Discussion

The result of this paper is a geometry, and a single sharp test of it. The geometry: trained attention lives on the projective null cone, the query–key dot product is a null-ray inner product, the conformal dimension is a homogeneity degree on the cone, and layer depth is RG flow toward the conformal vacuum of an eternal two-dimensional black hole. The test: the raw query–key score is log-linear in token distance, and its slope is the conformal dimension at $\rho = 0.976$ — confirming the null-ray reading at the level of the mechanism, not merely the ensemble.

We are deliberately precise about the seams. The literal coefficient of the log-distance law is renormalized by the softmax (the raw slope is $\approx -2.8\Delta$, not -2Δ); we report the rank correspondence as robust and the absolute coefficient as normalization-dependent. Log-linearity is a universal property of the positional geometry, not a mark of the conformal heads; the conformal dimension is the selective quantity, and it is this that sits at the SYK fixed point. The boundary-corrected power law on a finite sequence is derived and confirmed to fit the position-dependence of attention, with a boundary coefficient positive almost everywhere—the direction of the attention sink. The specific BCFT parametrization of that correction (the image form ($\dagger$)) is phenomenological: an adversarial model comparison (exp-060, pre-registered) shows the exponential boundary layer outperforms it at matched complexity. The boundary effect is real; the image-form identification of its structure is not yet established.

Robustness (Phase 0 checks). Before submission we ran four gatekeeping tests. (W1) Split-half stability: Δ and valley measured on disjoint input halves for Pythia-410m and Pythia-1.4b; cross-half $\rho \approx 0.67\text{--}0.81$, inside the pre-registered keep band; full closure requires Pythia-2.8B (pending, exp-059). (W3) Conditioning-artifact exclusion: the joint statistic $P(\Delta \in [0.20, 0.30] \text{ AND } R^2 \geq 0.90)$ is 0.09 for GPT-2 and 0.000 in all randomized-weight and shuffled-profile null replicates ($p = 0.001$, both protocols); pooled cross-model density excess $p \approx 10^{-20}$ (exp-058). (W4) λ -sign resolution: the verbal prediction (positive $\rho(\lambda, \text{valley})$) was wrong; deriving the exact V-shaped valley functional recovers the negative sign from first principles, and the model-implied valley $\hat{v}(\hat{\Delta}, \hat{\lambda})$ predicts the behavioral statistic at $\rho = 0.68\text{--}0.91$ with no free parameters (exp-061). None of these qualifications weakens the central claim; each sharpens it.

What “convergence to the causal structure of light” means, made concrete: the conformal group is the symmetry of the null cone; the deep layers of a trained transformer organize their attention

to sit at the fixed point of that symmetry; and the quantity they compute is an inner product of light-cone representatives of positions in the sequence. The geometry is not imported as a metaphor. It is the geometry the measurements have been pointing at.

Reproducibility

All code and per-head data are in the public repository `Capacity-For-Evil/ariel`:

- exp-057 (BCFT image-form derivation test): `research/physics/experiments/exp-057_bcft_image_lambda`
- exp-056 (log-distance test): `research/physics/experiments/exp-056_qk_log_distance/`
- exp-055 (layer-resolved Δ , attention entropy): `research/physics/experiments/exp-055_delta_attention`
- exp-060 (adversarial model comparison, boundary form): `research/physics/experiments/exp-060_bcft_a`
- exp-058 (null distributions, conditioning-artifact exclusion): `research/physics/experiments/exp-058_delt`
- exp-059 (split-half stability): `research/physics/experiments/exp-059_split_half_stability/`
- exp-061 (λ -sign derivation): `research/physics/experiments/exp-061_lambda_sign_derivation/`
- exp-046/047/048/049 (GOE universality, untrained controls): `research/physics/experiments/`
- The geometric assembly note: `development/status/rooms/study/notes/2026-06-09_attention_lives_o`

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